Nicolas Bigeard

MSc candidate | System 2 reasoning: search, verification and recurrent depth

Seeking a 6-month research internship starting February or March 2027 | **NeurIPS 2026 workshop poster**

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Research Interests

I want to work on **System 2 reasoning**: what governs a model's decision to keep pursuing a line of reasoning or drop it, what extra test-time computation actually buys, and how much thinking happens as continuous internal computation in recurrent depth. My results measure the baseline first. Search over Lean proofs buys at most 2.1 points over stock tactics with no model, and for tied-depth students I prove a threshold above which searching the sharing schedule can't be justified by its estimation cost.

Education

ESILV, École Supérieure d'Ingénieurs Léonard de Vinci Paris, France
MSc (diplôme d'ingénieur), Financial Engineering & Quantitative Finance *Sept. 2023 – 2027*

- Year 3: *Félicitations du jury*, every module 15.00–18.64/20. Year 4: research modules 17.33 and 18.41/20. 2025–26: research gap year.
- Coursework: Advanced Probability, Advanced Statistics, Stochastic Processes, Machine Learning.

Classes Préparatoires aux Grandes Écoles, MPSI / MP La Roche-sur-Yon, France
Lycée Saint François d'Assise. National selective track in mathematics and physics *Sept. 2021 – June 2023*

- **Ranked 1st of 24 in MP in both semesters**; 1st in mathematics (semester 2). MPSI: 4th, then 3rd of 24.

Research Experience

Centre Borelli, ENS Paris-Saclay Paris, France
Research Intern, continuing as collaborator | With A. Kalogeratos *May 2026 – Present*

- Derived the first out-of-sample extension for multi-view diffusion operators: exact on the training set and training-free.
- Built the family's first inductive benchmark (19 datasets, Holm-corrected over 17 comparisons). The searched operator leads published competitors by +0.10 AMI and ties the equal-weight baseline; a learned encoder can do no better than the truncated SVD it imitates, at about 300× the cost.

Paul Wurth S.A. Luxembourg
Research Intern, Causal Discovery | Supervised by C. Schockaert | preprint *Apr. 2025 – Feb. 2026*

- First author of **Tucker-CAM**, non-linear causal discovery for industrial time series that scales to 2,889 variables.

SauronID
Co-founder & CEO | sauronid.eu *Feb. 2026 – Present*

- Authorization and verifiable-audit gateway for AI agents: checks the mandate a human owner signed, writes tamper-evident receipts.

Publications & Preprints

Measure the Library First: What a Formalization Pipeline's "Verified" Output Actually Asserts | *sole author* 2026 | preprint | **Accepted, NeurIPS 2026 AI for Science Workshop (poster)**

- Audited my own arXiv-to-Lean 4 pipeline: of twenty theorems it reported as machine-verified, one carries mathematical content. Twelve close with a single tactic over constant stubs; one uses an axiom that is the theorem itself. Traced six paths to "proved" without a proof, each with a fix and a regression test.
- On miniF2F, twenty Mathlib tactics with no model close 36.1% of statements against 26.0% for a full retrieval-augmented search loop, so search adds at most 2.1 points (95% confidence). That floor moves 7.2 points in one Mathlib window.

Two Costs of Tying Depth: Matching Minimax Rates and When Each One Binds | *sole author* | preprint 2026

- Matching minimax bounds for how much a recurrent-depth student should share across depth, $\asymp \min\{r^2, \frac{\sigma^2}{nL}(mP + L\log m)\}$, via Assouad for the cores and Fano over sharing patterns.
- A regime threshold $P^\star = L\log m/m$: whole-block tying exceeds it by seven orders of magnitude, so searching arbitrary sharing schedules can't pay for itself. Tested on nine pretrained teachers (GPT-2, Pythia); the mixed outcome is reported as such.

Nash Initialization for Recurrent-Depth Transformers | *sole author* | preprint 2026

- Stability of System 2 recurrent-depth transformers over many iterations without normalisation: LayerNorm applies a per-cycle rank-$(d-2)$ projection, and $O(1/T)$ residual scaling is the critical exponent.

Technical Skills

Mathematics: minimax theory, model selection, spectral and diffusion geometry, Markov operators, convex optimisation, Lean 4
Languages & tools: Python, C++, Lean 4, OCaml; PyTorch, CUDA, NumPy, Polars, scikit-learn, Docker, Linux, Git, LaTeX
Methods: Monte-Carlo tree search, RL for reasoning, causal structure learning, experimental design with multiple-comparison correction

Additional

Also: game-theoretic interpretability of chess engines, supervised by Y. Vlassopoulos (IHES)
Languages: French (native), English (fluent), Spanish (B1). **Leadership**: Co-founder & Head of Research, EuroTech Federation; TEDx PULV team
Competitions: Agent Hackathon (Hugging Face, Anthropic); Iterate Paris (Anthropic). **Athletics**: 2nd, French nationals; Ironman